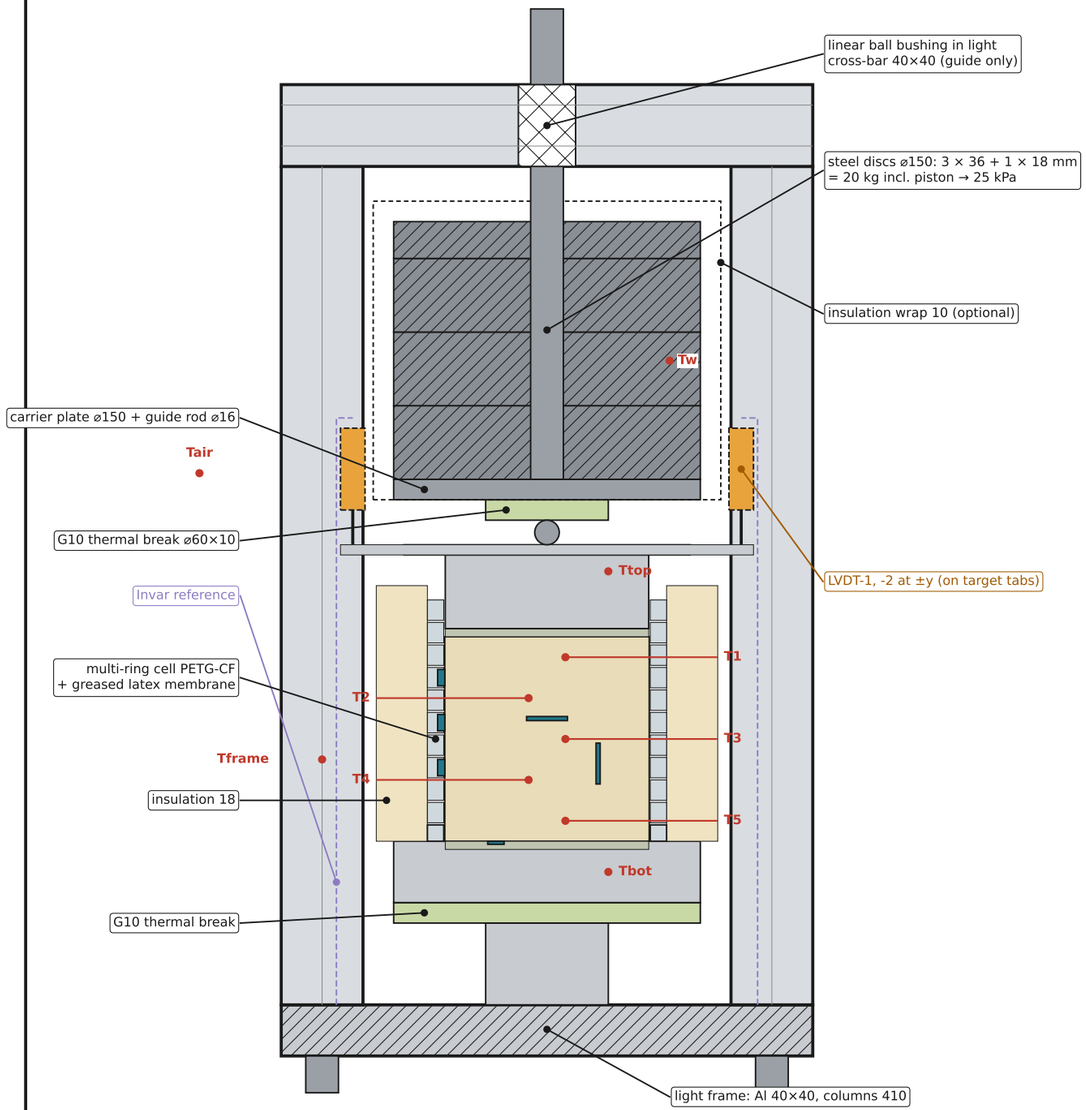


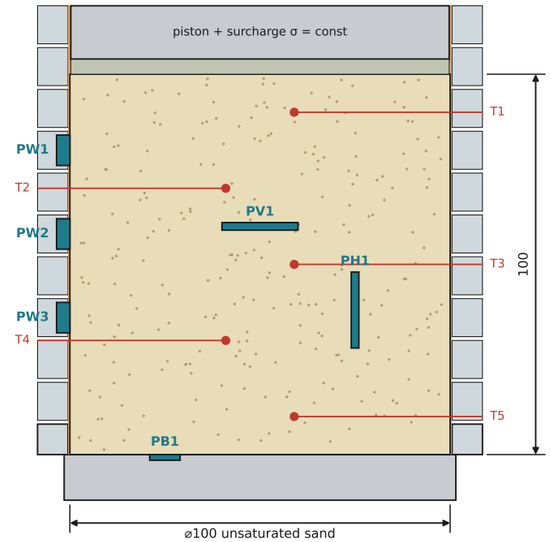
Sensors (tag → model → CR1000)

tag	measures	recommended model	CR1000
PV1 / PH1	vert. / radial stress	TML PDA-PA 200 kPa on ø20×2 disc	MUX-1 (4×16), BrFull, VX3 2 V
PB1	vertical stress at base	TML PDA-PA 200 kPa (flush)	MUX-1 (4×16), BrFull, VX3 2 V
PW1…PW3	lateral stress on wall	TML PDA-PA 200 kPa (flush)	MUX-1 (4×16), BrFull, VX3 2 V
LVDT-1,2	check: no movement	RDP DCTH400	SE7–10, VoltSE ±5 V
T1…T5	soil, r 9 mm	Campbell 109SS	MUX-2 → SE13, VX2
Ttop, Tbot, Tw	piston, pedestal, weights	Campbell 109SS	(same)
Tframe, Tair	frame, air	Campbell 109SS / 107	MUX-2 → SE13, VX2

σ kPa	total kg	full discs	half discs	trim kg
5	4.0	0	0	1.25
10	8.0	1	0	0.33
15	12.0	1	1	1.88
20	16.0	2	1	0.96
25	20.0	3	1	0.04

How it works
Gravity gives an exactly constant surcharge; fixed part ≈ 2.8 kg ≈ 3.4 kPa (piston, carrier, rod).
No load cell, no adjustment; the frame only guides the rod and carries the LVDTs.
Weights lag the freezer air by ≈ 0.6 K at the 0.1 K/h ramp (vypocty.md §7).

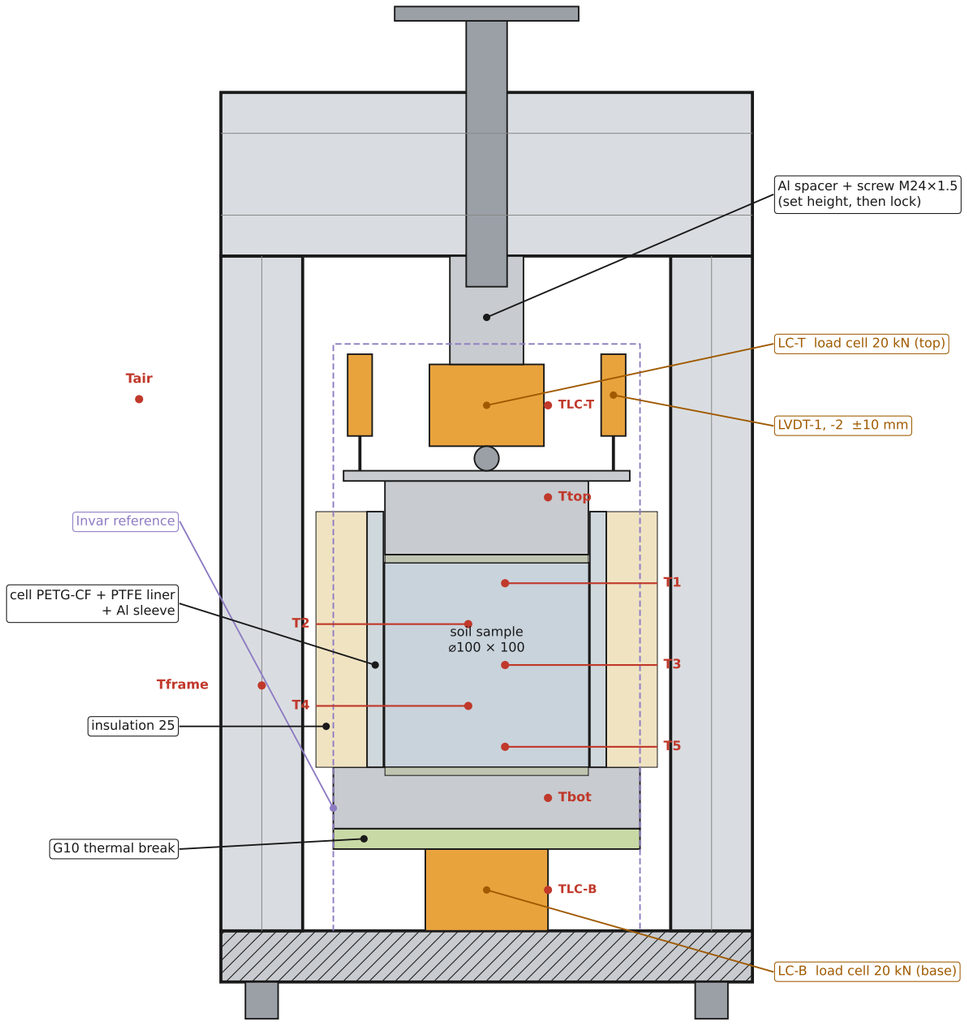
DETAIL – sand, pressure sensors, thermistors ≈ 1 : 1.6



schematic; orange = displacement, blue = pressure, red = temperature; cells and LVDTs revolved into section

Variant A – unsaturated sand, dead-weight surcharge ≤ 25 kPa

status	DRAFT	drawn	Claude Code (draft)	checked: —
units	mm	project	E7 – freeze–thaw loading tests	date 2026-10-20
scale	≈ 1 : 3	drawing no.	**E7-V01**	sheet 1 / 6



Sensors (tag → model → CR1000)

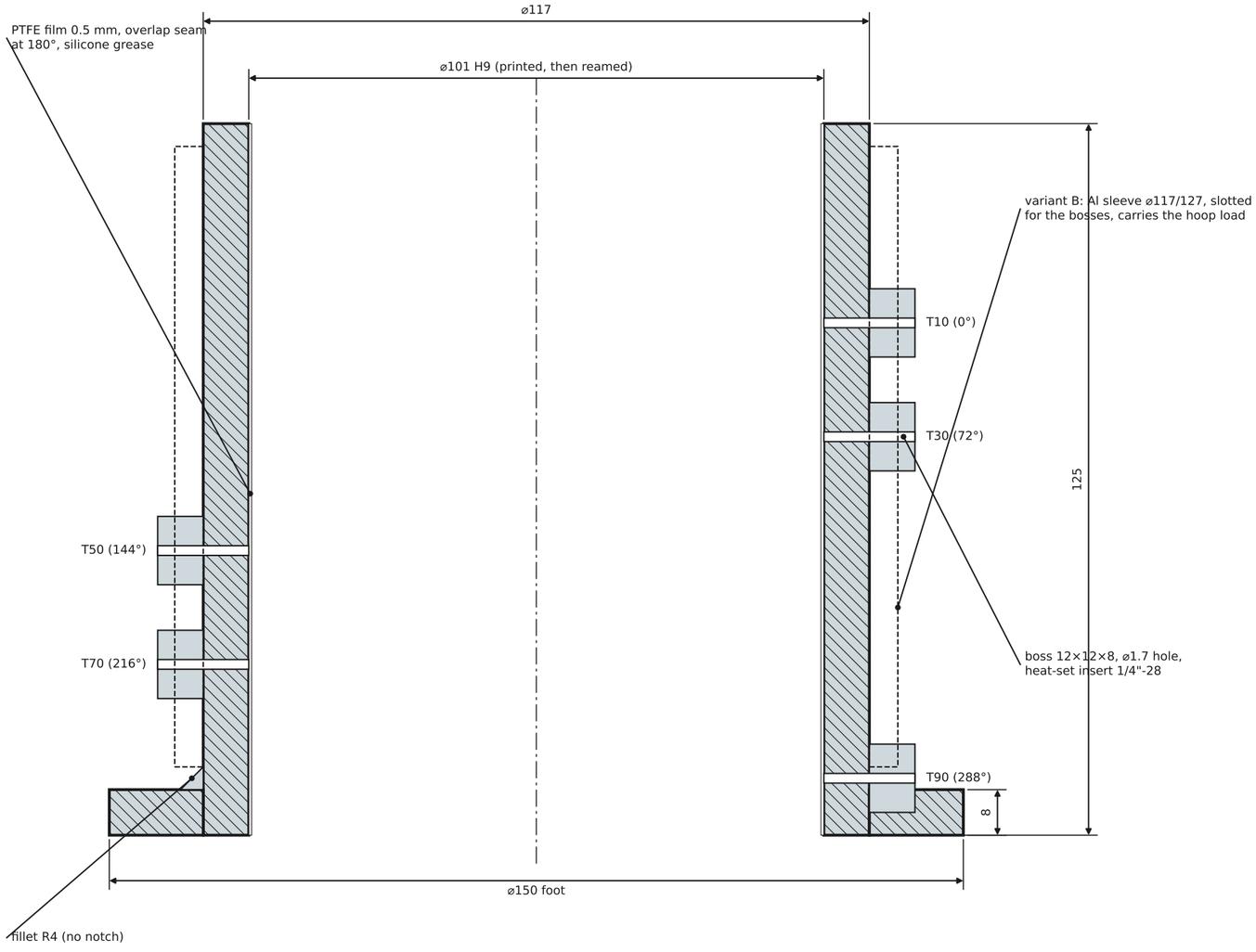
tag	measures	recommended model	CR1000
LC-T	heave force, top	HBM C9C/20kN	DIFF1, BrFull, VX1
LC-B	force at base	HBM C9C/20kN	DIFF2, BrFull, VX1
LVDT-1,2	residual movement	RDP DCTH400	SE7–10, VoltSE ±5 V
T1…T5	soil, r 9 mm	Campbell 109SS	MUX-2 → SE13, VX2
Ttop, Tbot	piston, pedestal	Campbell 109SS	(same)
TLC-T/B, Tframe, Tair	load cells, frame, air	Campbell 109SS / 107	MUX-2 → SE13, VX2

How it works
The screw holds the piston at constant height; the soil pushes against it while freezing. LC-T measures that force, LC-B the force arriving at the base – the difference is the friction on the cell wall. The LVDTs show how much the frame still gives (report the stiffness with the pressure).
Run a blank test with an Al dummy to remove the effect of the frame shrinking in the cold (vypocty.md §3).

schematic, proportions to scale; sensors in orange (force, displacement) and red (temperature)

Variant B – constant height (simplified)

status	DRAFT	drawn	Claude Code (draft)	checked: —
units	mm	project	E7 – freeze–thaw loading tests	date 2026-10-20
scale	**≈ 1 : 3**	drawing no.	**E7-V02**	sheet 2 / 6



Material check – PETG-CF

load case	stress	SF short	SF long
hoop stress, A 200 kPa	1.3 MPa	> 30	> 9
hoop stress, B 1 MPa	6.3 MPa	6–9	1.9–2.6
hoop stress, B 2 MPa	12.6 MPa	3–4	1.0–1.3
hoop, B 2 MPa + Al sleeve 5 mm	≈ 1 MPa*	> 40	> 12
axial (Z), friction μ 0.5, 1 MPa	5.8 MPa	2.6–5	< 1.6
axial (Z), PTFE μ 0.05, 1 MPa	0.6 MPa	> 25	> 7

Assumed strength (data-sheet range, verify): XY 40–55 MPa, Z 15–30 MPa, E 3–6 GPa; long-term limit ≈ 30 % (creep, layer bond). Details: vypocty.md §4.

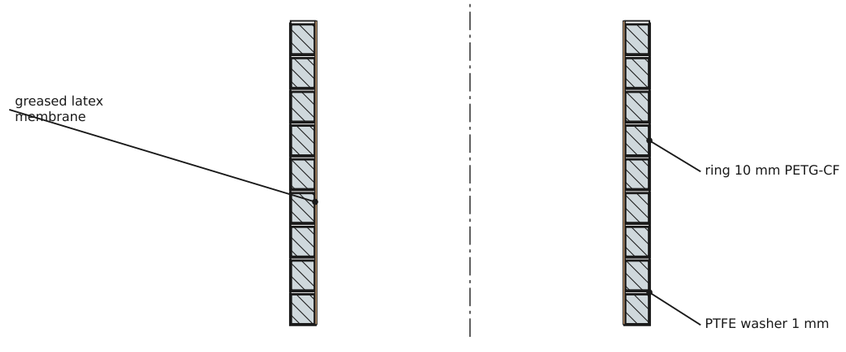
Verdict: suitable for variant A; for B (design 2 MPa) only with the Al sleeve shown dashed (*it carries ≈ 90 % of the hoop load: 70 GPa×5 mm vs. ≈ 4 GPa×8 mm), the PETG-CF then acts as a liner.
Conditions for both:
- the wall is ≥ 8 mm and printed with concentric perimeters only (100 % infill, no zig-zag), so the hoop stress runs along the filament;
- friction is kept low (PTFE liner) and the cell is not clamped at the foot, so friction pushes the wall instead of pulling the layers apart;
- the inside is sealed (thin epoxy coat) so water cannot enter print voids and split layers when it freezes;
- there are no printed threads or sharp notches (CF lowers toughness at −10 °C) – heat-set inserts and fillets instead.

Variant A: a multi-ring cell (view F) is the literature standard against wall friction – printed rings 10 mm + PTFE washers + greased latex.

Print settings (starting point)
hardened nozzle 0.6 mm, layer 0.2 mm, 250–260 °C, bed 80 °C, slow outer walls, filament dried 4 h at 65 °C, enclosure, part cooling ≤ 30 %.

Qualification before use
1 ring burst test: printed ring 20 mm high, internal water pressure to failure (target ≥ 3× design pressure);
2 10 freeze–thaw cycles −10/+10 °C of water-saturated coupons, then tensile test XY and Z vs. dry control;
3 creep: ring under 1 MPa for 7 days at +10 °C, hoop strain gauge.

VIEW F – multi-ring cell (variant A) 1 : 2

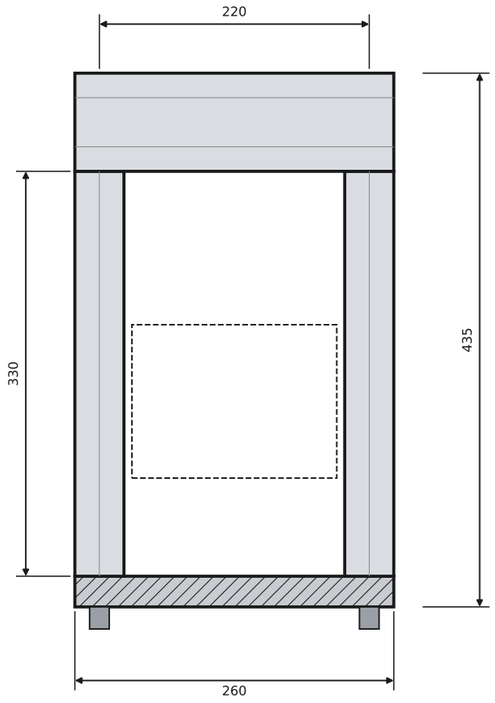


print upright; hoop direction = layer direction; foot rests on the pedestal, not bolted

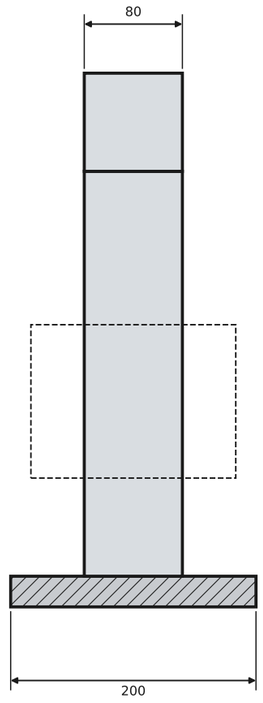
Sample cell – PETG-CF, 3D printed

status	DRAFT	drawn	Claude Code (draft)	checked: —
units	mm	project	E7 – freeze–thaw loading tests	date 2026-10-20
scale	**1 : 1**	drawing no.	**E7-V03**	sheet 3 / 6

FRONT VIEW 1 : 5



SIDE VIEW 1 : 5



Bill of materials

#	part	size	qty
1	base plate Al 6082 T6	260 × 200 × 25	1
2	profile 40×80 (B) / 40×40 (A) column	l = 330 (B) / 410 (A)	2
3	profile 80×80 (cross-beam)	l = 260	1
4	core bolt M12 into tapped profile end		8
5	socket screw M10 base → column end		4
6	levelling foot M12, steel		3
7	Invar rod ⌀10 + Invar/Al bridge (LVDT)	l = 300	2 + 1
8	linear ball bushing ⌀16 (A) / nut M24×1.5 (B)	in cross-beam	1
9	end caps, T-slot covers		–

Profile values used in vypocty.md §2 are typical catalogue values (series 8): 40×80 A ≈ 1050 mm², 80×80 I ≈ 1.4·10⁶ mm⁴ – check against the chosen supplier.

Stiffness at 15.7 kN (2 MPa): cross-beam 0.036 mm, columns 0.035 mm, load cell ≈ 0.05 mm, joints 0.02–0.15 mm. Joints dominate → core bolts into tapped profile ends, torque-controlled, no angle brackets; optionally 2 steel tie rods M12 next to the cell.

Fit in the chest freezer: frame 260 × 200, height 435 mm incl. base; variant B; variant A uses a light 40×40 frame, columns 410 mm (it only guides the weights); two rigs side by side on the tray – layout and heights on E7-V06.

one frame for both variants; core-bolted joints, no angle brackets in the load path

Frame – aluminium profiles

status	DRAFT	drawn	Claude Code (draft)	checked: —
units	mm	project	E7 – freeze–thaw loading tests	date 2026-10-20
scale	**1 : 5**	drawing no.	**E7-V04**	sheet 4 / 6

Options per measurement (PICK = recommended)

sensor	maker / model	key specs	verdict
LC 20 kN	HBM C9C/20kN	⌀26×13 mm, 1 mV/V, nom. −10…+70, op. −30…+85 °C	PICK
LC 20 kN	Burster 8402 20 kN	1.5 mV/V, ≤ 50 µm deflection, op. −30…+100 °C, IP54	alternative
LC 20 kN	HBM C10 25 kN	class 0.02–0.05, height 36 mm [U]	alternative
LC 20 kN	Futek LTH500 (≈22 kN)	donut, 0.05 mm, op. −50…+93 °C	alternative
LC 20 kN	Interface LBM	comp. only +21…+77 °C	not suitable
LVDT ±10	RDP DCTH400	±5 V / 0–10 V out, −40…+80 °C, ±0.25 % FS	PICK
LVDT ±10	Micro-Epsilon DTA-10D + MSC7401	0–5 V / 4–20 mA, −20…+80 °C	alternative
LVDT ±10	Solartron AX/10/S + OD5	±5 V out, needs 230 V conditioner	alternative
LVDT ±10	HBM WA-L 20 mm	passive, needs carrier amplifier	alternative
LVDT ±10	TE DC-EC 500	operating only 0…+70 °C	not suitable
p cell	TML PDA-PA 200 kPa	⌀6.5×1 mm, 0.8 mV/V, comp. −10…+60, op. −20…+70 °C	PICK
p cell	Kyowa BEC-A 200/500 kPa	⌀30 mm, for model tests; t, temp. [U]	one cell only
p cell	TML KDE/KDF-PA	⌀50 mm – too large for ⌀100 sample	not suitable
p wall	TE/MEAS EPL	miniature, comp. −40…+90 °C, ⌀ [U]	alternative
p cell	Tekscan FlexiForce A201	thin film, drift < 5 %/log t	comparison only
T probe	Campbell 109SS	⌀1.6 mm, ±0.38 °C at 0 °C, direct on CR1000	PICK
T probe	Campbell 107	⌀7.6 mm, ±0.2 °C (0–50 °C)	air / frame
T probe	TE/YSI 44031 bead	±0.1 °C, own half bridge needed	for 0.01 °C (E6 chain)
T probe	Pt100 1/10 DIN + 4WPB100	2 DIFF + VX per sensor	too many channels

CR1000 channel plan

terminal	connected to	measurement
DIFF1–2	LC-T, LC-B (rig B)	BrFull, reversal; VX1 2500 mV
SE7–SE10	LVDT-A1, A2, B1, B2	VoltSE ±5000 mV, averaging; own ±15 V supply
DIFF6 (SE11/12)	MUX-1 AM16/32B, 4×16	6 pressure bridges (PV1, PH1, PB1, PW1–3), BrFull
VX3	MUX-1 excitation	2000 mV (TML: < 2 V recommended)
C1, C2	MUX-1 reset, clock	
SE13	MUX-2 AM16/32B, 2×32	≈ 24 thermistors (Therm109), VX2
C3, C4	MUX-2 reset, clock	
12V, G	both multiplexers	
SE14–16	spare	

Channel count (2 rigs): 3 load cells, 4 LVDTs, 6 pressure bridges, ≈ 24 thermistors.
16 SE / 8 DIFF on the CR1000 → two AM16/32B: bridges (4×16), thermistors (2×32).

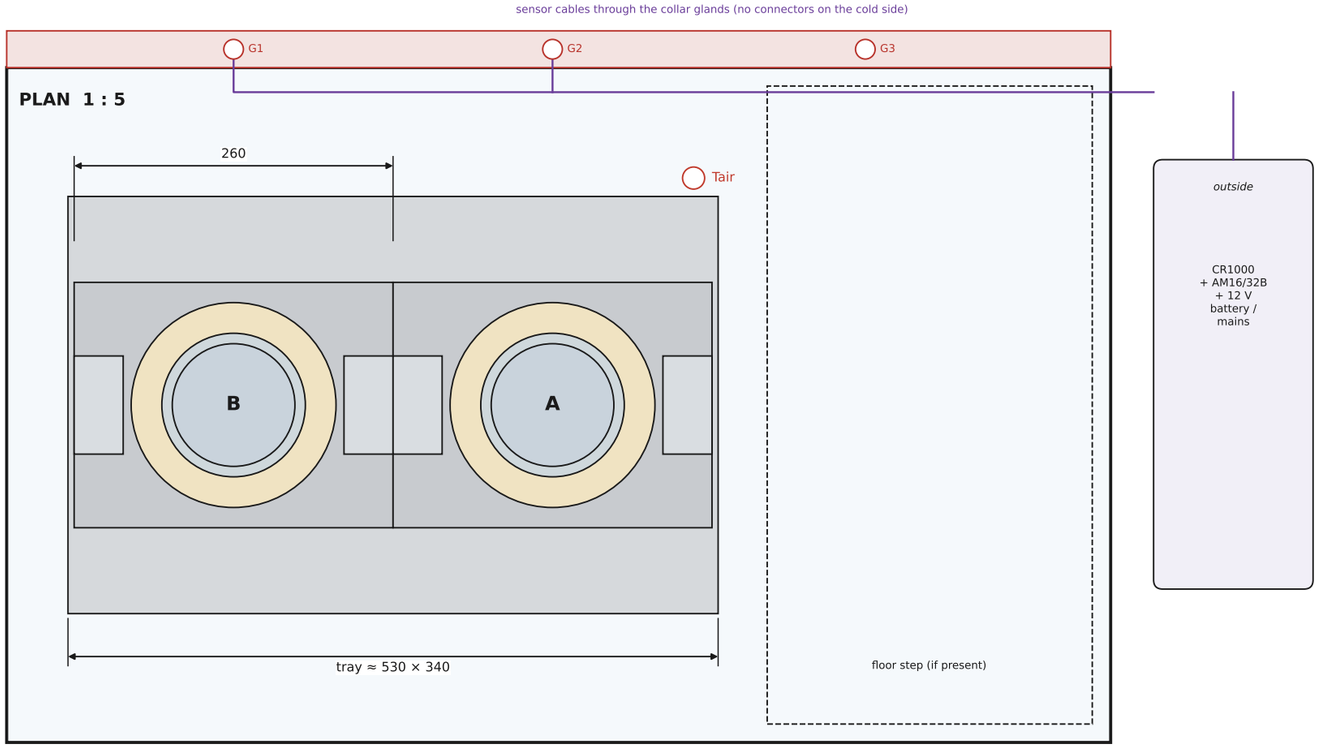
Resolution with the CR1000 (13-bit ADC)
- LC-T/B 1 mV/V: 2.5 mV FS → ≈ 2.6 N ≈ 0.3 kPa
- LVDT ±5 V over ±10 mm: ≈ 1.3 µm (0.7 µm with reversal); absolute linearity ±0.25 % FS ≈ ±50 µm → calibrate with a micrometer
- 109SS: ±0.38 °C at 0 °C; ice-point offset check improves it

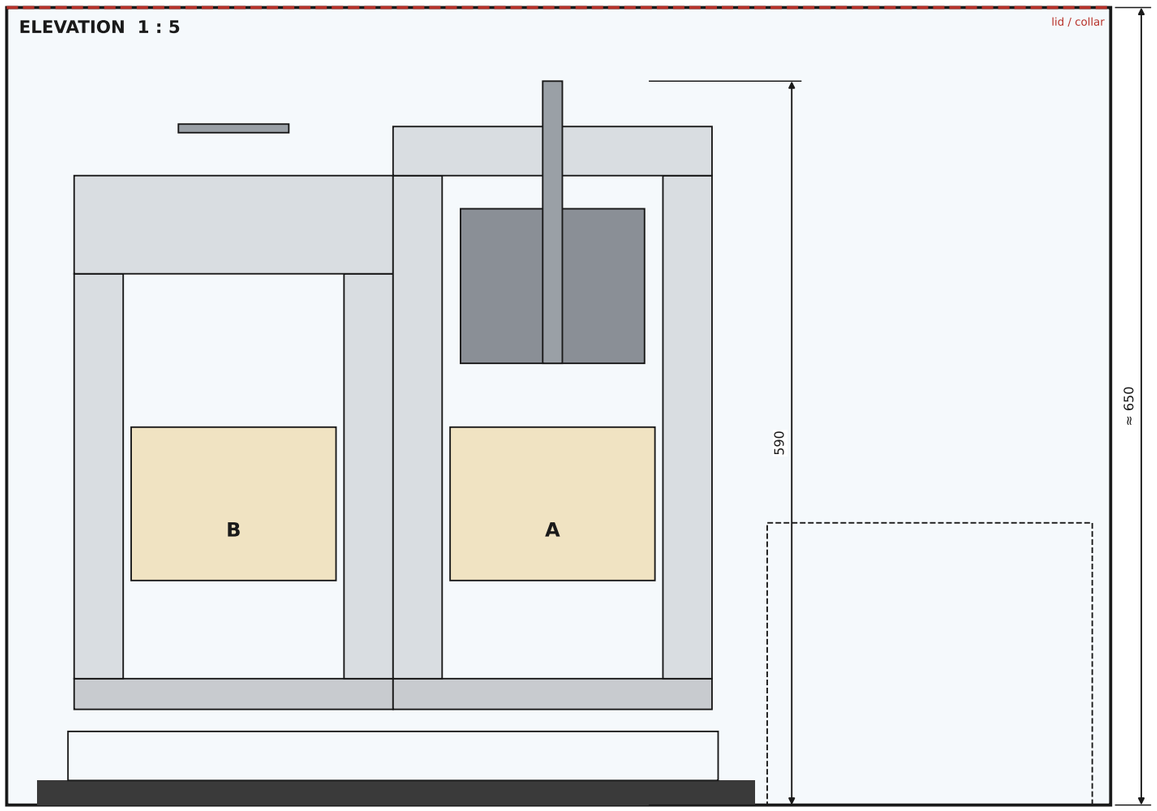
Notes
- CR1000 is discontinued (successor CR1000X / CR1000Xe, 24-bit); the existing one is fine for these resolutions.
- Logger outside the freezer (CR1000 −25…+50 °C).
- No load cell found with compensation to −20/−30 °C as standard: Interface SSM −15 °C, HBM −10 °C (op. −30 °C) – log TLC and correct; Interface/Lorenz offer extended ranges on request.
- 4WFBS modules are for quarter bridges – not needed here.
- Pressure cells: calibrate in the same sand under dead weight (25–200 kPa) at +10 and −10 °C; correct zero(T) with the neighbouring thermistor.

both rigs on one CR1000 + AM16/32B; models from manufacturer data (see README §7)

Sensor selection and CR1000 wiring

status	DRAFT	drawn	Claude Code (draft)	checked: —
units	mm	project	E7 – freeze–thaw loading tests	date 2026-10-20
scale	–	drawing no.	**E7-V05**	sheet 5 / 6





Fit check

item	size	limit	status
base plates 2 × 260 × 200	520 × 200	tray ≈ 530 × 340	OK (tight)
rig A (weights)	20 kg steel discs ø150	nothing outside frame	OK
highest point	590 mm	inside ≈ 650	OK, measure
logger	outside, 0…+40 °C	CR1000 −25…+50 °C	keep warm & dry

Notes
- Both rigs share the freezer program (Set A) and the CR1000; each rig has its own Tair sensor at sample mid-height and its own load cell temperatures.
- Both rigs are set from above with a screw and handwheel; the free end of the freezer stays empty (room for the cable bundle or a third rig).
- Run the blank test with Al dummies in both rigs before the first soil run.
- Dimensions marked ≈ are estimates from the photos (E6) – please measure.

same freezer as E6 (inside ≈ 900 × 550 × 650, to be measured); logger outside

Both rigs in the chest freezer + CR1000

status	DRAFT	drawn	Claude Code (draft)	checked: —
units	mm	project	E7 – freeze–thaw loading tests	date 2026-10-20
scale	**1 : 5**	drawing no.	**E7-V06**	sheet 6 / 6